27 28

PHYSICS HSSC–I

Inclusive Scheme of Studies 2024 (1200 Marks)

SECTION – A (Marks 17)

Time allowed: 25 Minutes

Candidate Sign. _____

Invigilator Sign. _____

Answer Sheet No. _____

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Version No.				
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ROLL NUMBER						

Deleting/overwriting is not allowed.
Do not use lead pencil.

Fill the relevant bubble against each question according to curriculum:

	Question	A	B	C	D	A	B	C	D
1.	Which of the following equations is dimensionally correct?	$v_f = v_i + at$	$v_f^2 = v_i^2 + 2at$	$s = v_i t + \frac{1}{2}at$	$F = ma^2$	○	○	○	○
2.	Pascal (Pa) is equal to:	N/m	N/m²	N²/m²	kg/m	○	○	○	○
3.	Time of flight of a projectile depends on:	Horizontal velocity	Mass	Vertical component of velocity	Acceleration only	○	○	○	○
4.	A car accelerates uniformly from rest to $60ms^{-1}$ in 5 seconds. What distance does it cover during this time?	150m	450m	300m	600m	○	○	○	○
5.	Which of the following is a correct way to represent Young's Modulus?	$\frac{F/A}{\Delta L/L}$	$\frac{F.A}{\Delta L}$	$\frac{F}{A.\Delta L}$	$\frac{F.\Delta L}{L.A}$	○	○	○	○
6.	In a steady, incompressible flow, if the cross-sectional area of a pipe decreases to one-third of its original size, what happens to the fluid velocity?	It decreases to one-third	It remains the same	It triples	It becomes nine times larger	○	○	○	○
7.	Direction of centripetal force is always:	Tangential	Perpendicular	Radially inward	Outward	○	○	○	○
8.	If the temperature of heat source is doubled than before, efficiency increases by:	1.5 times	2 times	2.5 times	3 times	○	○	○	○
9.	Antinode in a wave is point of:	Zero displacement	Maximum displacement	Zero velocity	No wave	○	○	○	○
1	What is the motion of an electron moving at an angle of 45° to a uniform magnetic field?	Linear motion	Circular motion	Parabolic motion	Helical motion	○	○	○	○

	Question	A	B	C	D	A	B	C	D
11.	A potentiometer is used to compare:	Resistance	Current	Potential difference	Power	○	○	○	○
12.	Electric field strength is:	Work done	Energy	Force per unit charge	Charge per force	○	○	○	○
13.	Postulates of special theory of relativity applies to:	Only light	Only motion	All physical laws	Only energy	○	○	○	○
14.	Correct Mass-energy relation is:	E = mv	E = mc²	E = mg	E = gh	○	○	○	○
15.	Proton is made up of:	2 down and 1 up quark	1 down and 2 up quarks	2 down and 2 up quarks	4 down and 2 up quarks	○	○	○	○
16.	If distance from axis increases, moment of inertia:	Decreases	Increases	Remains constant	Becomes zero	○	○	○	○
17.	Correct relation between torque and angular acceleration is:	$\tau = \frac{IR}{\alpha}$	$\tau = \frac{\alpha}{I}$	$\tau = \frac{I}{2}\alpha$	$\tau = m\alpha$	○	○	○	○

——1HA-I 26004-(30041) (B) ——

- $S = v_i t + \frac{1}{2}at^2$
- $I = \frac{1}{2}MR^2$
- $F_c = \frac{mv^2}{r}$
- $E = \frac{F}{q}$
- $V = \frac{W}{q}$
- $\eta = 1 - \frac{T_2}{T_1}$
- $I = \frac{2}{5}MR^2$
- $I = mr^2$
- $A_1v_1 = A_2v_2$
- $v_f = v_i + at$
- $E = mc^2$
- $Av = \text{constant}$

ROLL NUMBER						

PHYSICS HSSC–I

Inclusive Scheme of Studies 2024 (1200 Marks)

Time allowed: 2:35 Hours **Total Marks Sections B and C: 63**

SECTION – B (Marks 39)

Q. 2 Answer the following parts briefly. **(13 x 3 = 39)**

(i)	Deduce the dimensions of k from Coulomb's law $\left(F = k\frac{q_1 q_2}{r^2}\right)$.	03	OR	Describe any three properties of cross product of two vectors.	1x3
(ii)	Describe photon as a force carrier.	03	OR	Distinguish between the terms elastic deformation and plastic deformation.	03
(iii)	Show that: $\vec{A}.\vec{B} = A_x B_x + A_y B_y + A_z B_z$	03	OR	Show that ε and $\frac{\Delta\phi}{\Delta t}$ have the same units.	03
(iv)	Differentiate between conservative and non-conservative forces with examples.	03	OR	Distinguish between turbulent and streamline flow.	03
(v)	Calculate the resistance of wire 10 m long that has diameter of 3mm and resistivity of $2.63\times10^{-2}\Omega m$.	03	OR	A body of mass $(27.2\pm0.1)kg$ has a volume of $(2.0\pm0.1)m^3$. Calculate the density of the body? Find uncertainty in the density measurement.	03
(vi)	Derive the relationship $P.E = \frac{1}{2}\left[\frac{YA}{L}\right]x^2$	03	OR	In thermodynamics briefly describe: Reversible process Irreversible process Cyclic process	1x3
(vii)	A 5 kg shell is fired with a muzzle velocity of $700ms^{-1}$ from the gun barrel 2.4m long. Calculate the average value of: The acceleration The time to come out of the barrel	03	OR	Describe the working of Atomizers in perfume bottles.	03
(viii)	What is meant by precision? Also write the importance of taking multiple readings in an experiment.	1+2	OR	Find the area of the parallelogram whose adjacent sides are given by the vectors: $A = (\hat{i} + 6\hat{j} + 2\hat{k})m$ and $B = (7\hat{i} + \hat{j} + 5\hat{k})m$.	03
(ix)	Briefly explain light dependent resistor (LDR). Give one application of LDR.	2+1	OR	The front wheel of a tractor travels 700 revolutions while the rear wheel 280 in a time interval of 40 seconds. Find their angular velocities.	03
(x)	Compare constructive and destructive interference.	1.5 +1.5	OR	Drive a relationship between torque and angular acceleration.	03
(xi)	State first law of thermodynamics. Briefly explain the term internal energy.	1.5 + 1.5	OR	What is meant by potential gradient? Derive its formula.	03
(xii)	Briefly explain the relationship between pressure and volume of a gas at constant temperature. Show it graphically as well.	2+1	OR	What is meant by superfluid? How is superfluid made?	1.5 + 1.5
(xiii)	Water flows through a 1cm diameter pipe with a speed of $1ms^{-1}$. What should be the diameter of the nozzle if the water is to emerge at $2.1ms^{-1}$?	03	OR	A stationary observer hears the sound of a car horn at a frequency that is 15% higher than the horn's actual frequency. If the speed of sound is $343ms^{-1}$, what is the speed of the car as it approaches the observer?	03

SECTION – C (Marks 24)

Attempt the following questions.

Q.3	a. Range of projectile and b. Height of projectile	3+3	OR	a. State postulates of special theory of relativity. b. Briefly discuss time dilation and length contraction. c. At what speed would the mass of the proton be tripled? The rest mass of a proton is 1.673×10^{-27} kg.	2+2 +2
Q.4	a. State Bernoulli's equation. b. Derive velocity of efflux using Torricelli's theorem. Draw diagram as well.	1+3 +2	OR	Derive an expression for a magnetic force experienced by a moving charge in a uniform magnetic field. Also discuss the direction of that force.	4+2
Q.5	a. Discuss working of a refrigerator. b. How increase in Entropy means degradation of energy? Explain with an example.	3+3	OR	a. State Kirchoff's laws for current and voltage. b. There is a copper coil having 2000 turns with $0.8mm^2$ cross-section area, the length per turn is 80cm. Calculate the resistance of the coil.	3+3
Q.6	a. Derive the expression for Kinetic energy. b. State work energy theorem in a resistive medium.	3+3	OR	Derive f_1, f_2 and f_3 for modes of vibration in a closed organ pipe.	2+2 +2

$F = G\frac{m_1 m_2}{r^2}$ $\quad \vec{A}.\vec{B} = AB\cos\theta$ $\quad KE = \frac{1}{2}mv^2$ $\quad \text{density} = \frac{\text{mass}}{\text{volume}}$ $\quad v = \sqrt{2gh}$ $\quad R = \frac{\rho L}{A}$

$PE = mgh$ $\quad \vec{A}\times\vec{B} = AB\sin\theta\hat{n}$ $\quad I = I_o\cos^2\theta$ $\quad A = \pi r^2$ $\quad P = \vec{F}.\vec{v}$ $\quad A_1v_1 = A_2v_2$

$c = 3\times10^8 ms^{-1}$ $\quad \text{Work} = \vec{F}.\vec{x}$ $\quad MV = mv$ $\quad v_f = v_i + at$ $\quad 2aS = v_f^2 - v_i^2$ $\quad v = r\omega$

$\omega = \frac{\theta}{t}$ $\quad N = \frac{S}{2\pi r}$ $\quad m = \frac{m_o}{\sqrt{1-\frac{v^2}{c^2}}}$ $\quad f_o = f_s\frac{v}{v \pm v_s}$ $\quad H = \frac{v_i^2\sin^2\theta}{2g}$ $\quad E = -\frac{\Delta V}{\Delta r}$

$\theta = 2\pi r$ $\quad \rho_{copper} = 0.02\mu\Omega m$

29 28

PHYSICS HSSC–I

Inclusive Scheme of Studies 2024 (1200 Marks)

SECTION – A (Marks 17)

Time allowed: 25 Minutes

Candidate Sign. _____

Invigilator Sign. _____

Answer Sheet No. _____

Signature should not cross this line

Version No.				
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ROLL NUMBER						

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Fill the relevant bubble against each question according to curriculum:

	Question	A	B	C	D	A	B	C	D
1	Which equation correctly relates torque (τ), moment of inertia (I), and angular acceleration (α)?	$\tau = I^2\alpha$	$\tau = \frac{I}{\alpha}$	$\tau = I + \alpha$	$\tau = I\alpha$	○	○	○	○
2	The dimensional formula of velocity is:	$[M^0L^0T^0]$	$[LT^{-1}]$	$[LT^{-2}]$	$[L^2T^{-1}]$	○	○	○	○
3.	Joule can be expressed as:	$kg\ m^{-2}\ s^{-2}$	$kg\ m\ s^{-1}$	$kg\ m^2\ s^{-1}$	$kg\ m^2\ s^{-2}$	○	○	○	○
4.	Horizontal velocity in projectile motion:	is zero at max height	increases	decreases	remains constant	○	○	○	○
5.	A bus travelling at $10ms^{-1}$ accelerates uniformly at $2ms^{-2}$. Calculate the velocity of the bus after 5 s.	$0.02ms^{-1}$	$20ms^{-1}$	$2.0ms^{-1}$	$0.2ms^{-1}$	○	○	○	○
6.	Which of the following is a correct way to represent Young's Modulus?	$\frac{F.\Delta L}{L.A}$	$\frac{F/A}{\Delta L/L}$	$\frac{F.A}{\Delta L}$	$\frac{F}{A.\Delta L}$	○	○	○	○
7.	Continuity equation is based on:	Force	Energy conservation	Momentum	Mass conservation	○	○	○	○
8.	If velocity of a body moving in circular path doubles, centripetal force becomes:	Four times	Same	Double	Half	○	○	○	○
9.	According to the first law of thermodynamics, if 200 Joules of heat is added to a system and 80 Joules of work is done by the system, the internal energy change is:	120J decrease	80J increase	120 J increase	280J increase	○	○	○	○
1	Stationary waves are formed by:	Diffraction	Reflection only	Superposition of waves	Refraction	○	○	○	○

II

	Question	A	B	C	D	A	B	C	D
11.	Path of charged particle in magnetic field is always:	Rondon	Straight	Parabolic	Circular	○	○	○	○
12.	At balance point, current in galvanometer is:	Zero	Maximum	Minimum	Infinite	○	○	○	○
13.	If 800J energy is used to move a charge of 4C between two points in an electric field. Potential difference is:	$5V$	$100V$	$200V$	$5\times0^{-3}V$	○	○	○	○
14.	The rest energy of mass 2.00g is:	$1.80\times10^{18}J$	$180\times10^{12}J$	$1.80\times10^{14}J$	$1.80\times10^{16}J$	○	○	○	○
15.	According to special theory of relativity time dilation means:	Time increases for moving observer	Time decreases for moving observer	Time becomes zero	Time is constant for moving observer	○	○	○	○
16.	Neutrino is:	Magnetic particle	Charged particle	Heavy particle	Neutral particle	○	○	○	○
17.	A solid cylinder with mass M=5kg and radius R=0.3m rotates about its central axis. Calculate the moment of inertia of the cylinder.	$0.23kgm^2$	$5.3kgm^2$	$4.7kgm^2$	$1.5kgm^2$	○	○	○	○

- $v_f = v_i + at$
- $I = \frac{1}{2}MR^2$
- $F_c = \frac{mv^2}{r}$
- $E = \frac{F}{q}$
- $V = \frac{W}{q}$
- $E = mc^2$
- $I = \frac{2}{5}MR^2$
- $I = MR^2$

ROLL NUMBER						

PHYSICS HSSC–I

Inclusive Scheme of Studies 2024 (1200 Marks)

Time allowed: 2:35 Hours **Total Marks Sections B and C: 63**

SECTION – B (Marks 39)

Q. 2 Answer the following parts briefly. **(13 x 3 = 39)**

(i)	Deduce the dimensions of universal gravitational constant (G). $\left(F = G\frac{m_1 m_2}{r^2}\right)$	03	OR	Describe any three properties of dot product of two vectors.	1x3
(ii)	Give Comparison of amorphous and polymeric solids. (Three points)	1x3	OR	State the Higgs Boson as a fundamental particle which is responsible for the particle' s mass.	03
(iii)	The magnitudes of dot and cross product of two vectors are $6\sqrt{3}$ and 6 respectively. Find the angle between the vectors.	03	OR	How a seismometer uses electromagnetic induction to detect earthquakes? Explain briefly.	03
(iv)	In a science experiment a 3.00 kg water rocket, is launched from ground. The rocket's total energy at the top of its flight is 2352 J. a. What was the rocket's launching speed? b. What height did the rocket went?	03	OR	Find the power delivered by the engine for attaining velocity (3,4) ms^{-1} while it exerts a force (8, -2) N.	03
(v)	Calculate the resistance of a wire 10 m long that has diameter of 2 mm and resistivity of $2.63\times10^{-2}\Omega m$.	03	OR	Differentiate between accuracy and precision of measurements. (Give a graphical example)	03
(vi)	Briefly explain brittle and ductile materials. (In context of elastic and plastic deformation)	1.5 +1.5	OR	State first law of thermodynamics. Discuss its significance.	1+2
(vii)	Evaluate effect of air resistance on projectile motion. Show graphically.	2+1	OR	Briefly explain Magnus effect (an application of Bernoulli Effect). Draw diagram as well.	2+1
(viii)	Write any three limitations of dimensional analysis.	03	OR	Briefly discuss any three properties of an ideal fluid.	1x3
(ix)	What is magnetic flux? Also describe magnetic flux density and write its formula.	1+2	OR	Write any three characteristics of stationary waves.	1x3
(x)	What are beats? How are they produced?	1+2	OR	Determine the equivalent value of 1 radian in degrees.	03
(xi)	Briefly describe: a. Unit cell and its types b. Crystal lattice	2+1	OR	Derive the relation between Torque and angular acceleration.	03
(xii)	What is a light dependent resistor (LDR)? Why is LDR useful in automatic circuits?	1+2	OR	Explain the use of beats in tuning musical instruments.	03
(xiii)	The orbit of an electron revolving around the nucleus of an atoms is considered to be circular of radius 50 pm and the period of motion is 0.2 fs. Find out angular velocity of electron (ω).	03	OR	The electric field at a point due to a point charge is 26 N/C and the electric potential at that point is 13 J/C. Calculate the distance of the point from the charge.	03

SECTION – C (Marks 24)

Attempt the following questions.

Q.3	a. What is meant by polarization. b. Discuss one application of polarization. c. A polarized light of intensity I_o is passed through another polarizer whose axis makes an angle of 60° with the axis of the polaroid filter. What is the intensity of transmitted polarized light from second polaroid filter?	1+2 +3	OR	a. State Archimedes principle. b. A water storage tank in a residential area develops a small leak near its lower end. The top of the water in the tank is 10m above the point of the leak. (i) With what speed does the water rush from the hole? (ii) If the hole has an area of $0.080cm^2$, how much water flows out in one second?	1+5
Q.4	Describe elastics collision in one dimension. Derive relation to show the relative velocities before and after collision.	1+5	OR	Briefly explain Elastic Potential Energy. Derive its mathematical formula.	2+4
Q.5	State working principle of Carnot's Engine. With the help of graph discuss its all four processes in detail.	1+1 +4	OR	A ball of 100gm is thrown vertically upward at a speed of 25m/s. If no energy is lost, determine the height it would reach. If the ball only rises to 25m, calculate the work done against the air resistance. Also calculate the force of friction.	06
Q.6	a. What is meant by term protentional gradient? Derive its formula. b. Show that $\frac{volt}{meter} = \frac{newton}{columb}$	3+3	OR	a. State postulates of special theory of relativity. b. Briefly discuss length contraction as a consequence of special relativity. c. The length of spaceship is 100m long. What will be its length if the spaceship moves with a speed of 0.99c?	2+1 +3

$F = G\frac{m_1 m_2}{r^2}$ $\quad \vec{A}.\vec{B} = AB\cos\theta$ $\quad KE = \frac{1}{2}mv^2$ $\quad \text{Work} = \vec{F}.\vec{x}$ $\quad v = \sqrt{2gh}$ $\quad R = \frac{\rho L}{A}$

$PE = mgh$ $\quad \vec{A}\times\vec{B} = AB\sin\theta\,\hat{n}$ $\quad I = I_o\cos^2\theta$ $\quad A = \pi r^2$ $\quad P = \vec{F}.\vec{v}$ $\quad A_1v_1 = A_2v_2$

$c = 3\times10^8 ms^{-1}$ $\quad \text{density} = \frac{\text{mass}}{\text{volume}}$ $\quad MV = mv$ $\quad v_f = v_i + at$ $\quad 2aS = v_f^2 - v_i^2$ $\quad v = r\omega$

$\omega = \frac{\theta}{t}$ $\quad N = \frac{S}{2\pi r}$ $\quad m = \frac{m_o}{\sqrt{1-\frac{v^2}{c^2}}}$ $\quad f_o = f_s\frac{v}{v \pm v_o}$ $\quad H = \frac{v_i^2\sin^2\theta}{2g}$ $\quad E = -\frac{\Delta V}{\Delta r}$

$\theta = 2\pi r$ $\quad \rho_{copper} = 0.02\mu\Omega m$

Version No.				
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ROLL NUMBER						

PHYSICS HSSC–I
(Curriculum 2022-23) (OLD)
SECTION – A (Marks 17)
Time allowed: 25 Minutes

Candidate Sign. _____

Invigilator Sign. _____

Answer Sheet No. _____

Deleting/overwriting is not allowed.
Do not use lead pencil.

Fill the relevant bubble against each question according to curriculum:

	Question	A	B	C	D	A	B	C	D
1.	The dimensions of torque are:	$[M^{-1}L^{3}T^{-2}]$	$[ML^{2}T^{-2}]$	$[M^{2}L^{-1}T^{-2}]$	$[ML^{2}T^{-1}]$	○	○	○	○
2.	If the percentage error in mass is 1% and velocity is 2% then the possible percentage error in momentum is:	3%	4%	5%	8%	○	○	○	○
3.	For which angle the equation $\|A \times B\| = \|A.B\|$ is correct?	30°	45°	60°	76°	○	○	○	○
4.	The F_x component of a force $\vec{F}$ of magnitude 50N making an angle of 60° with X-axis, is:	25 N	5 N	15 N	10 N	○	○	○	○
5.	Two balls of mass 10 kg and 2 gm, collide head-on elastically. If 2 gm ball is stationary and 10 kg ball is moving at 2 m/s before the collision, what is the velocity of 2 gm ball after collision?	0 m/s	4 m/s	3 m/s	6 m/s	○	○	○	○
6.	A projectile has maximum range at:	90°	60°	45°	30°	○	○	○	○
7.	1 radian =	67.3°	60°	5.73°	57.3°	○	○	○	○
8.	If mass of a moving body is doubled but velocity is reduced to one-fourth, then:	K.E increases 2 times	K.E remains the same	K.E decreases 8 times	K.E increases 4 times	○	○	○	○
9.	An ideal fluid has:	Variable velocity	High compressibility	Variable density	Zero viscosity	○	○	○	○
10.	Materials that do not undergo plastic deformation but suddenly break within Elastic region are called:	Amorphous	Ductile	Brittle	Polymers	○	○	○	○

	Question	A	B	C	D	A	B	C	D
11.	In an isochoric process:	Q = 0	Q = ΔW	Q = ΔU	Q = ΔU - ΔW	○	○	○	○
12.	Two tuning forks of frequencies 512 Hz and 516 Hz are sounded together. The number of beats produced per second is	512 Hz	516 Hz	1028 Hz	4 Hz	○	○	○	○
13.	Two charges are experiencing a force of 50 N when placed at 10 cm distance. If charges are placed at 5 cm distance, then force will be:	200 N	75 N	12.5 N	100 N	○	○	○	○
14.	The resistance of a certain length of wire of area 8 mm^2 is 30Ω. If the area is changed to 24 mm^2, then new resistance is:	10 Ω	30 Ω	40 Ω	50 Ω	○	○	○	○
15.	The Time Period of cyclotron is given by:	$T = \frac{qB}{2\pi m}$	$T = \frac{qB}{4\pi m}$	$T = \frac{2\pi m}{qB}$	$T = \frac{qB}{4m}$	○	○	○	○
16.	A passenger passes an electricity pole placed along the side of a road with a speed $\frac{c}{2}$. The length of electricity pole observed by him is:	$l = l_\circ$	$l = \frac{\sqrt{3}}{2} l_\circ$	$l = \frac{2}{\sqrt{3}} l_\circ$	$l = \frac{2}{\sqrt{5}} l_\circ$	○	○	○	○
17.	Proton is made up of:	2 down and 1 up quark	1 down and 2 up quarks	2 down and 2 up quarks	4 down and 2 up quarks	○	○	○	○

- $p = mv$
- $A.B = AB\cos\theta$
- $\overline{A} \times \overline{B} = \hat{n}AB\sin\theta$
- $KE = \frac{1}{2}mv^2$
- $R = \frac{\rho L}{A}$
- $F = \frac{Kq_1q_2}{r^2}$
- For $m_1 >> m_2$; $u_2 = 0$ $v_2 = 2u_1$

ROLL NUMBER						

PHYSICS HSSC–I
(Curriculum 2022-23) (OLD)

Time allowed: 2:35 Hours **Total Marks Sections B and C: 68**

SECTION – B (Marks 42)

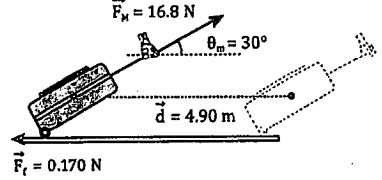
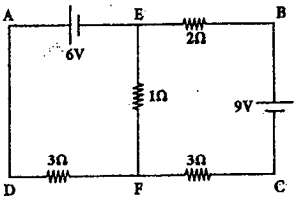
Q. 2 Answers the following parts briefly. **(14 x 3 = 42)**

(i)	If voltage measured across a conductor is $V \pm \Delta V$ = (7.3 ± 0.3) volts and current, is I±ΔI = (2.73 ± 0.051) ampere. Find the resistance and uncertainty in it?	03	OR	A peddler is pushing a trolly on a horizontal road with a force of 50 N making 30° with the road. Find the horizontal and vertical components of force?	1.5 + 1.5
(ii)	Prove that the given equation $v_f = f_i + at$ is dimensionally correct.	03	OR	If $\|A \times B\| = 3/4$ and $\|A.B\| = 1/2$ then find the angle between A and B?	03
(iii)	Derive a mathematical relation for height of projectile.	03	OR	Determine horizontal and vertical component of velocity of a projectile when launched horizontally.	03
(iv)	What is relation between linear and angular acceleration? Derive a mathematical expression.	03	OR	What is relation between Torque and Angular Momentum? Derive a mathematical expression.	1+2
(v)	Deriver the formula for kinetic energy.	03	OR	Explain work-energy theorem in a resistive medium.	1+2
(vi)	Using Archemedes principle, prove that upthrust is equal to weight of the liquid displaced. (Diagram is not required)	03	OR	Discuss the working of aerofoil.	03
(vii)	Calculate the strain energy stored in a 1m long steel wire connected to the roof, holding a weight of 980N. The diameter of steel wire is 0.004m and $\Delta L = 0.00039m$.	03	OR	A copper wire has a length of 50 m and cross-sectional area of 1.5 x 10^{-6} m^2. The resistivity of copper is 1.68 x 10^{-8} Ω m. Calculate the resistance of the wire?	03
(viii)	Using Ideal Gas equation verify that PV = NkT	1.5 +1.5	OR	Apply Kinetic Theory of Gases to prove that ρ =PM/RT	03
(ix)	If amplitude of a wave is doubled how much energy is increased?	03	OR	How is magnetic field created around a straight current carrying conductor? How its direction is determined?	2+1
(x)	An electric field has a strength of 5 N/C. What force is exerted on a charge of 2 C placed in this field?	03	OR	Prove that $E = -\frac{\Delta V}{\Delta r}$	03
(xi)	What is meant by the elastic potential energy of a material? Derive a mathematical relation for it.	1+2	OR	Using diagram derive formula for unknown resistance in Wheatstone Bridge.	1+2
(xii)	A particle with charge -3e moves at a velocity of 1 x 10^6 m/s at an angle of 45° and experiences a magnetic force of 6.72 x 10^{-14} N. What is the magnetic field strength?	03	OR	The frequency of a certain string is 125 Hz, when it vibrates in 5 segments. Calculate the frequencies when the string is vibrating in 3 and 4 segments. What is the speed of the wave in it, if its length is 0.80m?	03
(xiii)	What is Mass-Energy Equivalence Relation? Explain it briefly.	1+2	OR	At what speed would the mass of proton be tripled? The rest mass of proton is 1.673 x 10^{-27} kg.	03
(xiv)	Distinguish between α, β and γ particles on the basis of their charge and mass?	1x3	OR	State the Higgs Boson as a fundamental particle which is responsible for the particle's mass.	03

SECTION – C (Marks 26)

Attempt the following questions.

(Use of graph paper is not necessary. Candidates can make their own grid on answer book if required)

Q.3	Explain Carnot Engine and Carnot Theorem with the help of its graph. Find the relation for its efficiency.	4+3	OR	a. Describe the Doppler's effect for a moving source and a stationary listener. b. Discuss any two applications of Doppler's effect.	4+3
Q.4	Explain how electric and magnetic fields can be used in velocity selection. Make diagram as well.	4+2	OR	State Coulomb's Law. Write its mathematical expression. Two point charges $q_1 = 4\mu C$ and $q_2 = 2\mu C$, exert a repulsive force of 36N on each other. Calculate the distance between the charges.	1+1 +4
Q.5	A girl is dragging a suitcase and pulls it a distance of 4.90 m along level ground. She is applying a constant force of 16.8N at 30° with the horizontal. A 0.170 N friction force opposes the suitcase's motion. Find **(a)** work done by Girl **(b)** work done by frictional force and **(c)** net work done by all the forces acting on the suitcase? 	07	OR	a. Derive the formula for combined resistance of two or more resistors in series. b. Calculate the current that flows in the 1Ω resistor in the following circuit. 	3+4
Q.6	Determine f_1, f_2 and f_3 for stationary waves in an open pipe.	2x3	OR	How is artificial gravity created to counter weightlessness. Determine linear velocity angular velocity, time period and frequency of artificial satellite.	2+4

- $V = IR$
- $I = \frac{P}{A}$
- $F = \frac{kq_1q_2}{r^2}$
- $1eV = 1.6 \times 10^{-19} J$
- $\overline{A}.\overline{B} = AB\cos\theta$
- $A = \pi r^2$
- $E = \frac{F}{q}$
- $\overline{A} \times \overline{B} = AB\sin\theta$
- $f_n = nf_1$
- $v = f\lambda$
- $v_f = v_i + at$
- $v_f^2 - v_i^2 = 2aS$
- $F_c = \frac{mv^2}{r}$
- $P.E = \frac{1}{2}\left[\frac{YA}{L}\right]x^2$
- $R = \frac{\rho L}{A}$
- $F = qrB\sin\theta$
- $Y_{steel} = 20 \times 10^{10} N/m^2$

Version No.				
5	9	0	4	1

ROLL NUMBER						

PHYSICS HSSC–I

(Curriculum 2006) (OLD)

SECTION – A (Marks 17)

Time allowed: 25 Minutes

Candidate Sign. _____

Invigilator Sign. _____

Answer Sheet No. _____

Deleting/overwriting is not allowed.
Do not use lead pencil.

Fill the relevant bubble against each question according to curriculum:

	Question	A	B	C	D	A	B	C	D
1.	$\frac{1km}{1Gm} =$	k	μ	M	n	○	○	○	○
2.	If 20N force is acting at 60° with horizontal. $F_x =$	20N	15N	10N	5N	○	○	○	○
3.	What is the angle between $\vec{A}$ and $\vec{B}$ for which $\lvert\vec{A}+\vec{B}\rvert = \lvert\vec{A}-\vec{B}\rvert$?	30°	45°	60°	90°	○	○	○	○
4.	A graph is drawn with force along Y-axis and displacement along X-axis. The area under the graph represents:	Impulse	Couble	Moment of force	Work	○	○	○	○
5.	The time rate of change of momentum gives:	Force	Impulse	Acceleration	Power	○	○	○	○
6.	The energy stored in the spring of watch is:	K.E	Electrical Energy	Elastic P.E	Solar Energy	○	○	○	○
7.	Which of the following instruments works on the principle of moments?	Physical balance	Spring balance	Measuring cylinder	Vernier caliper	○	○	○	○
8.	Which one is constant for a satellite in orbit?	Velocity	K.E	Angular Momentum	Potential Energy	○	○	○	○
9.	Work is said to be negative when $\vec{F}$ and $\vec{d}$ are:	Parallel	Anti Parallel	Mutually Perpendicular	At 45°	○	○	○	○
10.	The speed of efflux is equal to the speed gained by fluid while falling through height h under the action of gravity, is statement of:	Equation of continuity	Torricelli's theorem	Bernoulli's theorem	Venturi's theorem	○	○	○	○
11.	In simple harmonic motion (S.H.M):	$a \propto \frac{1}{x}$	$a \propto m$	$a \propto -\frac{1}{x}$	$a \propto -x$	○	○	○	○
12.	In order to double the period of a simple pendulum:	Its length should be doubled	Its length should be quadrupled	The mass of its bob should be doubled	The mass of its bob should be quadrupled	○	○	○	○
13.	Two tuning fork of frequency 256 Hz and 258 Hz are sounded together. Number of beats observed are:	5	2	514	516	○	○	○	○
14.	Drag force, $F_d =$	$6\pi\eta v$	$\frac{6\pi\eta}{rv}$	$6\pi r$	$6\pi\eta rv$	○	○	○	○
15.	The principle of Michelson interferometer is based on the division of:	Wavefront	Amplitude	Frequency	Speed of light	○	○	○	○
16.	Two bodies are said to be in thermal equilibrium if they have the same:	Temperature	Amount of heat	Specific heat	Thermal capacities	○	○	○	○
17.	For an ideal gas system, the internal energy is directly proportional to:	Pressure	Volume	Mass	Temperature	○	○	○	○

• $T = 2\pi\sqrt{\frac{\ell}{g}}$ • $\vec{A}.\vec{B} = AB\cos\theta$ • $\vec{A}\times\vec{B} = AB\sin\theta\hat{n}$ • $W = F.d\cos\theta$

PHYSICS HSSC–I
(Old Curriculum 2006)

Time allowed: 2:35 Hours **Total Marks Sections B and C: 68**

SECTION – B (Marks 42)

Q. 2 Answers the following parts briefly. **(14 x 3 = 42)**

(i)	Determine the equivalent value of 1 radian in degrees.	03	OR	A bus horn produces sound of frequency 200 Hz and is receding from a stationary observer at 18 m/s. if speed of sound is 331 m/s then what frequency does the observer hear?	03
(ii)	Show that the equation for liquid pressure is $P = pgh$ dimensionally correct.	03	OR	Briefly explain the working of RADAR.	03
(iii)	The magnitude of dot and cross product of two vectors are $6\sqrt{3}$ and 6 respectively. Find the angle between the vectors.	03	OR	An oil film spreading over a wet footpath shows colors. How does it happen? Explain.	03
(iv)	Briefly explain with example, that for a body in complete equilibrium both first and second conditions of equilibrium must be satisfied.	03	OR	State the two rules of Huygens's principle.	03
(v)	Write any two applications of polarized light.	03	OR	Is it possible to convert internal energy into mechanical energy? Explain with an example.	03
(vi)	Briefly explain impulse and state how it is related to linear momentum.	03	OR	Briefly explain Isothermal process.	03
(vii)	If two balls of equal masses have elastic collision in one dimension, calculate velocities after collision.	03	OR	Briefly explain vector product of 2 vectors. Give one example.	2+1
(viii)	A ball of mass 100 g is thrown vertically upward at a speed of $25ms^{-1}$. If no energy is lost, determine the height it would reach.	03	OR	Briefly explain angular velocity and angular acceleration. Also their formulas.	1.5 + 1.5
(ix)	Write any three characteristics of waves.	03	OR	Derive an expression of velocity of a Disc when it rolls down an inclined plane.	03
(x)	A body will be weightless when the elevator falls down just like a free falling body. Explain.	03	OR	With the help of one daily life example define damped oscillations.	03
(xi)	Why the terminal speed of a parachutist after opening a parachute is less to the terminal speed before opening a parachute?	03	OR	The speed sound in air at $0°C$ is $332ms^{-1}$. What will be the speed of sound at $220°C$.	03
(xii)	Velocity of water in 6 inch diameter pipe is $5fts^{-1}$. find the velocity in a 3 inch diameter pipe which connects with it.	03	OR	The length of a pendulum is $(100 \pm 0.1)cm$. If acceleration of free fall is $(9.8 \pm 0.1)ms^{-2}$. Calculate percentage in time period of the pendulum.	03
(xiii)	What is torque? Explain why it is equal to the vector product of force and moment arm.	03	OR	Does the tension in the string of a swinging pendulum do any work? Explain.	03
(xiv)	Differentiate between free and forced oscillations with example.	03	OR	Smoke rises in a chimney when a breeze is blowing. Use Bernoulli's principle to explain the formula.	03

SECTION – C (Marks 26)

Attempt the following questions.

Q.3	a. What is meant by absolute potential energy? Derive a relation for absolute potential energy of body at the surface of the earth. b. A uniform rod 1m long with weight 6 N can be supported in a horizontal position on a sharp edge with weights of 10 N and 15 N suspended from its ends. What is the position of point of balance.	4+3	OR	What is Young's double slit experiment? Discuss in detail with the help of a diagram. Also find expressions of position of fringes on screen and fringe spacing.	07
Q.4	Define centripetal force and derive its relation.	06	OR	Keeping in mind the relative velocity of source and observer discuss Doppler effect in detail also compare real and apparent frequencies.	06
Q.5	a. Explain Newton's formula for the speed of sound. Show that how it was corrected by a French scientist Laplace? b. A Carnot heat engine absorbs 2000 J of heat from the source of heat engine at 227°C and rejects 1200J of heat during each cycle to sink. Calculate efficiency of engine, temperature of sink and amount of work done during each cycle.	4+3	OR	a. What is projectile motion? Give examples. Derive mathematical equations for range of a projectile. b. Calculate the angle of projection for which K.E. at the highest point of its trajectory is equal to one-fourth of its K.E. at point of projection.	4+3
Q.6	Describe viscous drag. Also derive an expression for terminal velocity.	06	OR	Discuss the effect of temperature and density on speed of sound in air.	3+3

- $R = \frac{v_i^2 \sin 2\theta}{g}$
- $H = \frac{v_i^2 \sin^2 \theta}{2g}$
- $T = \frac{2v_i \sin \theta}{g}$
- $f' = (\frac{v}{v+a})f$
- $f = \frac{1}{2\pi}\sqrt{\frac{g}{l}}$
- $P = \frac{mgh}{t}$
- $\vec{A} \times \vec{B} = AB \sin\theta \hat{n}$
- $v_t = \frac{2\rho g r^2}{9\eta}$
- $a_c = \frac{v^2}{r} = \frac{v^2}{R}$
- $F_c = \frac{mv^2}{r}$
- $T = 2\pi\sqrt{\frac{l}{g}}$
- $v_e = \sqrt{\frac{GM}{R+h}}$
- $\vec{A}.\vec{B} = AB \cos\theta$
- $T = 2\pi\sqrt{\frac{m}{k}}$
- $f' = \left(\frac{v-b}{v+a}\right)f$
- $d = \frac{1}{N}$
- $V_{esc} = \sqrt{2gR}$
- $P = Fv$
- $K.E = \frac{1}{2}mv^2$
- $d \sin\theta = n\lambda$
- $f' = \left(\frac{v+b}{v-a}\right)f$
- $v_t = v_0 + (0.61)t$
- $2d \sin\theta = m\lambda$
- $R_e = 6.4 \times 10^6 m$
- $P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2$
- $\sin^2\theta + \cos^2\theta = 1$
- $M_e = 6 \times 10^{24} kg$
- $G = 6.673 \times 10^{-11} Nm^2 kg^{-2}$